YEAR 10 MATHS · CHAPTER 6 · MIND MAP

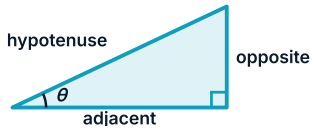
# Trigonometry

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## 1 Trigonometric ratios

SOH CAH TOA.

Hypotenuse 12, angle 50°: opposite $= 12\sin 50^\circ \approx 9.19$




## 2 Finding unknown angles

Use the inverse buttons: $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$.

$\cos\theta = \frac{5}{8} \Rightarrow \theta = \cos^{-1}(0.625) \approx 51.3^\circ$

## 3 Applications in 2D

Elevation and depression are measured from the horizontal.

From a 40 m cliff, the angle of depression to a boat is 25°

Distance $= \dfrac{40}{\tan 25^\circ} \approx 85.8$ m

## 4 Directions and bearings

True bearings: clockwise from north, three digits. Back bearing: add or subtract 180°.

N30°E = 030°T  S40°W = 220°T

Back bearing of 060°T is **240°T**

## 5 Applications in 3D

Find the right triangle in the plane you need, often using Pythagoras first.

Box 3 by 4 by 12: base diagonal 5

Angle with the base: $\tan\theta = \frac{12}{5} \Rightarrow \theta \approx 67.4^\circ$

## 6 The sine rule

$\dfrac{a}{\sin A} = \dfrac{b}{\sin B}$: use it with a side and its opposite angle.

$A = 40^\circ,\ B = 65^\circ,\ a = 10$: $b = \dfrac{10\sin 65^\circ}{\sin 40^\circ} \approx 14.1$

**Watch out:** finding an angle can give two answers (the ambiguous case).

## 7 The cosine rule

$c^2 = a^2 + b^2 - 2ab\cos C$: use it with two sides and the angle between, or three sides.

$a = 7,\ b = 9,\ C = 60^\circ$: $c^2 = 49 + 81 - 63 = 67$, so $c \approx 8.19$

## 8 Area of a triangle

$A = \frac{1}{2}ab\sin C$, with $C$ between sides $a$ and $b$.

$a = 7,\ b = 9,\ C = 60^\circ$: $A = \frac{1}{2} \times 63 \times \sin 60^\circ \approx 27.3$

## 9 The unit circle

A point on the unit circle is $(\cos\theta, \sin\theta)$.

$\sin(180^\circ - \theta) = \sin\theta$  $\cos(180^\circ - \theta) = -\cos\theta$

$\sin 150^\circ = \frac{1}{2}$  $\cos 120^\circ = -\frac{1}{2}$

## 10 Graphs of trig functions

$y = \sin x$ and $y = \cos x$: period 360°, between −1 and 1.

$\sin x$ starts at 0, $\cos x$ starts at 1

$y = \tan x$: period 180°, asymptotes at 90° and 270°

## 11 Exact values and trig equations

$\sin 30^\circ = \frac{1}{2},\ \cos 45^\circ = \frac{\sqrt{2}}{2},\ \tan 60^\circ = \sqrt{3}$

Solve $\sin x = \frac{1}{2}$ for $0^\circ \le x \le 360^\circ$: $x = 30^\circ$ or $150^\circ$

**Watch out:** check every quadrant where the ratio has the right sign.